

Executive Summary

Ferry Capacity Utilization & Operational Efficiency Analytics

Jack Layton Ferry Terminal | Prepared for Toronto Government Parks, Forestry & Recreation

This analysis reviewed 10 years of ferry ticket data (2015–2025, over 261,000 fifteen-minute records) to answer one question: are ferry resources being deployed where and when they are actually needed? The findings point to a clear, actionable conclusion: congestion at the Jack Layton Ferry Terminal is driven overwhelmingly by season, not by day of the week — and current planning should reflect that.

94.3% of major congestion events happen in summer (Jun–Sep)	34.6% is summer's share of all operating days — congestion is wildly disproportionate	13 hrs longest single-day strain event, on Canada Day 2018	8.25% of all time intervals are classified as congested
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Key Findings

- Season is the dominant driver of congestion, not weekends. Summer months make up only 34.6% of all operating days but account for 94.3% of all major congestion events. Within summer, weekdays (52.9%) actually contribute slightly more congestion days than weekends (41.4%) — demand is spread across the whole week during peak season, consistent with tourism-driven rather than commuter-driven ridership.
- Holidays produce extreme, predictable single-day strain. The single longest sustained congestion event in the full 10-year dataset occurred on Canada Day 2018, lasting approximately 13 continuous hours from late morning through evening. Elevated activity around July 1st recurs in multiple years regardless of which weekday it falls on.
- Off-peak hours are not significantly over-staffed. Sustained idle periods make up less than half of one percent (0.46%) of all measured intervals, suggesting limited room for cost savings by simply cutting scheduled service — any efficiency review should focus on the structure of off-season hours rather than blanket reductions.
- Demand is highly volatile. Ferry activity swings sharply rather than following a steady, predictable pattern — a result of leisure and tourism-driven travel rather than regular commuter flow. This reinforces the need for flexible, season-wide staffing rather than fixed day-by-day schedules.

Recommendations

- Adopt season-long capacity planning strategies across the June–September period rather than concentrating resources primarily on weekends.
- Deploy additional vessels, staff, and crowd-management resources on statutory holidays, particularly Canada Day and other high-demand events.
- Focus cost-efficiency reviews on off-season scheduling structures rather than reducing service during active operating periods, where idle capacity is already minimal.
- Obtain official vessel capacity specifications to refine utilization calculations and improve future operational performance metrics.

- Continue monitoring extreme-demand intervals to support proactive planning for festivals, group bookings, and other exceptional demand events.

Overall, the analysis demonstrates that ferry system strain is largely a seasonal operational challenge rather than a weekly scheduling issue. The results provide a data-driven foundation for shifting from static calendar-based scheduling toward more flexible, season-focused resource planning that can improve passenger experience, operational efficiency, and service reliability.

Methodology Note

This analysis is based solely on ticket sales and redemption records and does not incorporate weather, special-event calendars, or external tourism data, which likely contribute to the patterns observed. It measures operational efficiency — how existing capacity is used — and is intentionally separate from demand forecasting. A full methodology, validated calculations, and supporting charts are available in the accompanying research paper and interactive dashboard.